



The Science of Food Cravings: How Our Brains Choose What We Eat

【BBC精选 | 美食诱惑背后的科学：大脑如何选择食物？】

Summary: Why do we crave certain foods like burgers and cakes? This video explores the neurological processes behind food choices, focusing on the brain's reward system. MRI studies show that the orbital frontal cortex lights up when we consume foods high in fat and sugar. External factors like food labels and social influences significantly alter our perception of taste. Understanding these mechanisms can help design healthier foods and improve dietary choices.

摘要：为什么我们对汉堡和蛋糕等食物情有独钟？本期视频深入解析食物选择背后的神经机制。研究发现，大脑眶额皮层会对高糖高脂食物产生强烈反应，而标签暗示和社会因素会显著改变味觉体验。实验表明，相同的食物因名称不同可引发截然不同的神经反应。了解这些机制有助于设计更健康的食品，并帮助人们做出更明智的饮食决策。





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🕒 Estimated Reading Time: 8 min.

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▶ Why do we like some foods more than others?

为什么我们偏爱某些食物胜过其他？

▶ What is it about burgers and cakes that we crave?

汉堡和蛋糕究竟有何魅力让我们如此#渴望#？

▶ And what is happening in our brains when we choose what to eat?

当我们选择食物时，大脑正经历着什么？

▶ If you hopped into an MRI machine and were offered a delicious chocolate milkshake,

若你躺进磁共振仪并被递上一杯#美味#的巧克力奶昔，

▶ you would see your brain's reward system light up like a fun fare.

会目睹大脑奖励系统如游乐场嘉年华般绚烂闪耀。

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▶ Right above your eyes is your orbital frontal cortex.

双眼正后方是您的#眶##额#皮层。

▶ A part of the brain which is especially developed in humans and primates.

这个脑区在人类和灵长类动物中尤为发达。

▶ Here, bundles of neurons respond to different sensations and nutrients.

这里成束的神经元对不同感官刺激和营养成分产生反应。

▶ Taste, smell, how smooth and rich the milkshake feels.

味觉、嗅觉、奶昔的顺滑与醇厚质感。

▶ And the more your neurons light up, the more delicious the food seems.

神经元激活越强烈，食物就显得越#美味#。

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▶ Two things which particularly delight these reward neurons are fat and sugar.

最令奖励神经元愉悦的当属脂肪与糖。

▶ Combinations of fat and sugar can be even more delightful,

脂肪与糖的结合尤显#诱人#，

▶ such as in that delicious milkshake or cakes or donuts.

譬如那杯#香醇#奶昔，或是蛋糕与甜甜圈。

▶ But your neurons don't just respond to these sensations.

但神经元不仅回应感官刺激。

▶ They also activate when you're planning what to eat,

它们也会在你规划饮食时#激活#，

▶ in a contest with each other to get you to choose them.

相互竞逐以左右你的选择。

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▶ Pick me, says silky smooth mouthfeel.

"选我"，丝滑口感轻声细语。

▶ No, me!

"不，选我！"

▶ Shout sugary sweetness.

糖分爆表的#甜腻感#高声叫嚷。

▶ You never picked me.

"你从不选我"。

▶ Quietly mumbles a smell of broccoli.

一缕#西兰花#的清香低声嘟囔。

▶ And once you have chosen the same neurons track your progress.

当你做出选择后，这群神经元会持续追踪进展。

▶ And as you eat, they get less and less active as you approach fullness.

随着进食接近饱足，它们的活跃度逐渐降低。

▶ But we're not entirely at the mercy of the demands of our orbital frontal cortex.

但我们并非完全受#眶##额#皮层需求的摆布。

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▶ Having information about the food can make a big difference.

掌握食物信息能带来显著改变。

▶ Hot back into that MRI machine and slurp down some soup.

重回磁共振仪，啜饮几口汤羹。

▶ There are two types.

汤有两种品类。

▶ One is labeled rich and delicious flavor, and the other is labeled boiled vegetable water.

一种标着"浓郁#鲜香#"，另一种写着"涮菜水"。

▶ Your neurons light up at the taste of rich and delicious, and less for boiled vegetable water.

神经元对浓郁#鲜香#汤反应热烈，对涮菜水则兴致缺缺。

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▶ But wait a minute, you've been tricked.

但稍等，你被蒙蔽了。

▶ Those were both the same soup.

这两份本是同一种汤。

▶ The only difference was the name, and that was enough to completely change your experience of it.

仅名称之差就足以彻底颠覆你的感官体验。

▶ This experiment was also done with wine, telling people that the wine costs more made their neuro-necativity increase, and the wine tasted better.

葡萄酒实验亦复如是：标价更高时，人们神经活跃度提升，酒味也显得更醇美。

▶ Another part of the brain involved in choosing food is the amygdala.

参与食物选择的另一脑区是杏仁核。

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▶ The amygdala also has a role to play when you choose where to go out with another person.

当你与他人相约外出就餐时，杏仁核同样发挥作用。

▶ If you've seen what they prefer in the past, your amygdala will have developed so-called simulation neurons.

若你了解对方过往偏好，杏仁核会培育出所谓模拟神经元。

▶ These predict the choice you think they will make, which you can then factor into your own suggestions of what to eat together.

这些神经元预测对方可能的选择，进而影响你提出的共餐建议。

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▶ Differences in our genes are also a factor

基因差异也决定了

▶ in how susceptible we are to the siren call of our reward neurons.

我们对奖励神经元#蛊惑#召唤的#易感性#。

▶ With some people being naturally more responsive to the reward we feel from eating sugar and fat than others,

有人天生对糖脂带来的奖励感更敏锐，

▶ scientific experiments give us clues about how our brains compute our choices of what to eat,

科学实验揭示了大脑如何#运算#饮食选择，

▶ but the way we experience these choices in our lives and in society is also complex.

但我们在生活与社会中实践这些选择的方式同样复杂。

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▶ Dr. Emily Kantoa, Assistant Professor of Media Studies, gives her take.

媒体研究助理教授艾米丽·坎托亚博士阐述其见解：

▶ We choose the food we eat for a lot of different reasons.

我们选择食物基于诸多不同缘由。

▶ What's available at the grocery store?

超市有哪些可选？

▶ What's convenient?

什么更方便？

▶ What's affordable?

什么更实惠？

▶ What do we have good memories about?

哪些承载美好回忆？

▶ What tastes good to us?

什么合我们口味？

▶ What do we think is healthy?

什么被我们认为健康？

 00:02:59

▶ What is our current health status?

当前健康状况如何？

▶ What defines our ideas about who we are?

什么定义了自我认知？

▶ So when we think about food in the digital age, one of the biggest things that's changed the way people eat and the kinds of foods that they're seeking out are social media platforms.

故在数字时代思考饮食时，社交媒体平台已成为颠覆人类进食方式与食物选择的最大推手之一。

▶ Instagram and the desire for people to be able to take beautiful food photos has transformed the idea that you are what you eat into, you are what you post.

Instagram与人们拍摄美食大片的渴望，将"人如其食"转变为"人如其帖"。

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▶ So we seek a lot of different things from the food that we eat.

因此我们从食物中寻求多种需求。

▶ We can seek comfort.

或寻慰藉。

▶ We can seek a connection to history, to our families, to our heritage.

或寻与历史、家族、#传统#的联结。

▶ But we can also seek in it a sense of control.

但亦可从中寻求掌控感。

▶ When we live in moments that are full of economic, political, and social strife,

当我们身处充满经济、政治与社会#纷争#的时代，

▶ sometimes we seek in food that sense of security and safety.

有时会通过食物追寻安全感。

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▶ So in those moments, we sometimes see people get really interested in ideas about naturalness, health,

故此间常见人们对天然、健康、

▶ and purity as a way to protect ourselves from contexts outside of our control.

#纯净#等概念的热衷，以此抵御不可控的外部环境。

▶ So food also tells stories about who we are, the full complexity of identity.

食物亦诉说着我们身份认同的完整#复杂性#。

▶ What we eat tells stories about our gender and our sexuality, our race and our ethnicity, our social class, or our aspirations about our social class, the region where we live, even whether we live in an urban or rural area.

饮食叙事关乎性别与性取向、种族与民族、社会阶级或阶级向往、居住地域乃至城乡差异。

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▶ What we eat tells these contradictory, complex stories about who we are.

食物讲述着这些矛盾而复杂的自我故事。

▶ In the future, we can use our knowledge of what's happening in our brains to design foods that are low in calories and are still attractive but healthy.

未来可借助大脑运作知识，设计低卡却诱人的健康食品。

▶ And we can help ourselves by understanding how our reward neurons plot to get what they want.

理解奖励神经元如何谋取所求，亦能助我们做出更好决策。

▶ We can be aware of times that we tend to make poor choices, like when we choose a food because of some label which appeals to us rather than because of its taste.

我们可警觉易做劣选的时刻，例如因吸引人的标签而非实际味道选择食物。

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▶ So in the end, we are at least not fully at the mercy of our reward neurons.

归根结底，我们至少不完全受奖励神经元的#支配#。

▶ We can use our understanding to help design healthy foods and make healthy choices.

可用认知成果助力设计健康食品，作出健康选择。

